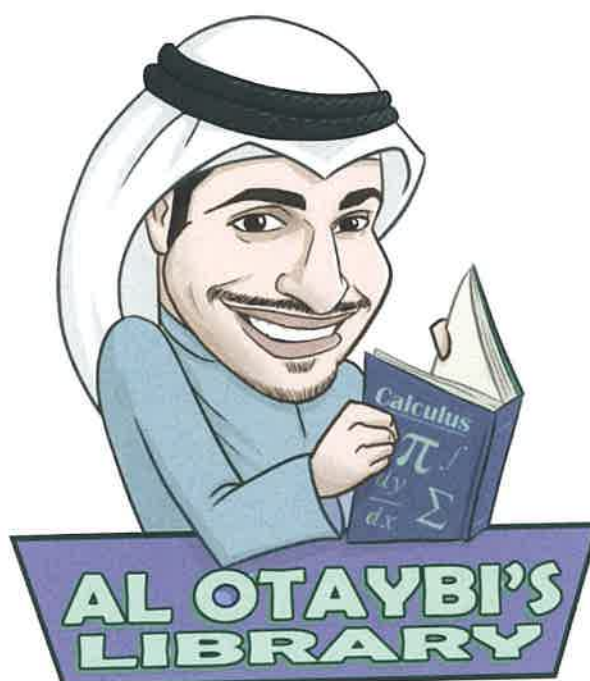


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 3.3 – 3.4

3.3 Derivatives of Trigonometric Functions.

3.4 The Chain Rule.

( 10 )



### 3.3 Derivative of trigonometric function



| Function ( الدالة ) | Derivative ( المشتقة ) |
|---------------------|------------------------|
| $\sin x$            | $\cos x$               |
| $\cos x$            | $-\sin x$              |
| $\tan x$            | $\sec^2 x$             |
| $\csc x$            | $-\csc x \cot x$       |
| $\sec x$            | $\sec x \tan x$        |
| $\cot x$            | $-\csc^2 x$            |



للتسهيل



— سالب

يكون فيها



$c$

الدوال المثلثية التي تبدأ بحرف

AL OTAYBI'S  
LIBRARY



## Text book questions



### Differentiate

$$f(x) = x^2 \sin x$$

اشتق بثنا + بثنا اشتق

$$f'(x) = 2x \sin x + x^2 \cos x$$

$$f(x) = e^x \cos x$$

اشتق بثنا + بثنا اشتق

$$f'(x) = e^x \cos x + e^x (-\sin x)$$

$$g(t) = t^3 \cos t$$

$$g'(t) = 3t^2 \cos t + t^3 (-\sin t)$$

$$h(\theta) = \csc \theta + e^\theta \cot \theta$$

اشتق بثنا + بثنا اشتق

$$h'(\theta) = -\csc \theta \cot \theta + e^\theta \cot \theta + e^\theta (-\csc^2 \theta)$$

$$y = \frac{x}{2 - \tan x}$$

$$y' = \frac{(2 - \tan x)(1) - x(-\sec^2 x)}{(2 - \tan x)^2}$$

$$y = \frac{\sin \theta}{1 + \cos \theta}$$

$$y' = \frac{(1 + \cos \theta) \cos \theta - \sin \theta (-\sin \theta)}{(1 + \cos \theta)^2}$$

## Differentiate

$$y = \sec x \tan x$$

استفاد بشتا + بشتا استفاد

$$y' = \sec x \tan x \tan x + \sec x \cdot \sec^2 x$$

$$y = \frac{\sec x}{1 + \sec x}$$

$$y' = \frac{(1 + \sec x) \sec x \tan x - \sec x \sec x \tan x}{(1 + \sec x)^2}$$

$$y = \frac{e^x \sin x}{1 + \tan x}$$

$$y' = \frac{(1 + \tan x) (e^x \sin x)' - e^x \sin x (1 + \tan x)'}{(1 + \tan x)^2}$$

$$= \frac{(1 + \tan x) (e^x \sin x + e^x \cos x) - e^x \sin x \sec^2 x}{(1 + \tan x)^2}$$

$$y = \frac{\cos x \sin x}{3 - \csc x}$$

$$y' = \frac{(3 - \csc x) (\cos x \sin x)' - \cos x \sin x (3 - \csc x)'}{(3 - \csc x)^2}$$

$$= \frac{(3 - \csc x) (-\sin x \sin x + \cos x \cos x) - \cos x \sin x (\csc x \cot x)}{(3 - \csc x)^2}$$

Find  $f'(x)$  where

$$f(x) = \frac{x^3 \sin x}{x^2 + 3}$$

$$\begin{aligned} f'(x) &= \frac{(x^2+3)(x^3 \sin x)' - x^3 \sin x (x^2+3)'}{(x^2+3)^2} \\ &= \frac{(x^2+3)(3x^2 \sin x + x^3 \cos x) - x^3 \sin x \cdot 2x}{(x^2+3)^2} \end{aligned}$$

Find  $f'(x)$  where

$$f(x) = \frac{x \sec x}{\sqrt{x} + 1}$$

$$\begin{aligned} f'(x) &= \frac{(\sqrt{x} + 1)(x \sec x)' - x \sec x (\sqrt{x} + 1)'}{(\sqrt{x} + 1)^2} \\ &= \frac{(\sqrt{x} + 1)((1) \sec x + x \sec x \tan x) - x \sec x \cdot \frac{1}{2\sqrt{x}}}{(\sqrt{x} + 1)^2} \end{aligned}$$

Find  $f'(x)$  where

$$f(x) = \frac{x^2 + 1}{e^x \sin x}$$

$$\begin{aligned} f'(x) &= \frac{e^x \sin x (x^2+1)' - (x^2+1)(e^x \sin x)'}{(e^x \sin x)^2} \\ &= \frac{e^x \sin x \cdot 2x - (x^2+1)(e^x(1) \sin x + e^x \cos x)}{(e^x \sin x)^2} \end{aligned}$$

Find  $f'(x)$  where

$$f(x) = \frac{x^3 \tan(x^2 + 1)}{1 - \sin x}$$

$$\begin{aligned} f'(x) &= \frac{(1 - \sin x) (x^3 \tan(x^2 + 1))' - x^3 \tan(x^2 + 1) (1 - \sin x)'}{(1 - \sin x)^2} \\ &= \frac{(1 - \sin x) [3x^2 \tan(x^2 + 1) + x^3 \sec^2(x^2 + 1) \cdot 2x] - x^3 \tan(x^2 + 1) (0 - \cos x)}{(1 - \sin x)^2} \end{aligned}$$

Find  $f'(x)$  where

$$f(x) = (x^2 - 1) \sec \sqrt{x-1} + \tan(x^3 - 1)$$

$$\begin{aligned} f'(x) &= 2x \sec \sqrt{x-1} + (x^2 - 1) \sec \sqrt{x-1} \tan \sqrt{x-1} \cdot \frac{1}{2\sqrt{x-1}} \\ &\quad + \sec^2(x^3 - 1) \cdot 3x^2 \end{aligned}$$

Find  $f'(\pi)$  where

$$f(x) = \sin x \tan x + \frac{3 \cos x}{1 + \sin x}$$

$$f'(x) = \cos x \tan x + \sin x \sec^2 x + \frac{(1 + \sin x)(-3 \sin x) - 3 \cos x \cos x}{(1 + \sin x)^2}$$

$$\begin{aligned} f'(\pi) &= 0 + 0 + \frac{0 - 3(-1)(-1)}{(1 + 0)^2} \\ &= \frac{-3}{1} = -3 \end{aligned}$$

**Remark**

$$\begin{aligned} \sin \pi &= 0 & \tan \pi &= 0 \\ \cos \pi &= -1 \end{aligned}$$



### Text book questions



If  $H(\theta) = \theta \sin \theta$ , Find  $H'(\theta)$  and  $H''(\theta)$ .

اشتقاق بشتا + بشتا اشتقاق

$$H'(\theta) = (1) \sin \theta + \theta \cos \theta$$

اشتقاق بشتا + بشتا اشتقاق

$$H''(\theta) = \cos \theta + (1) \cos \theta + (\theta) (-\sin \theta)$$

If  $f(x) = e^x \cos x$ , Find  $f'(x)$  and  $f''(x)$ .

اشتقاق بشتا + بشتا اشتقاق

$$\begin{aligned} f'(x) &= e^x \cos x + e^x (-\sin x) \\ &= e^x \cos x - e^x \sin x \end{aligned}$$

اشتقاق بشتا + بشتا اشتقاق

$$f''(x) = e^x \cos x + e^x (-\sin x) - [e^x \sin x + e^x \cos x]$$

If  $y = e^x \cos x$  , at  $(0, 1)$  Find

(1) Equation of the tangent line .

(2) Equation of the normal line .

اشتقاق  $y$  + اشتقاق  $\cos x$

$$y' = e^x (1) \cos x + e^x (-\sin x)$$
$$= e^x \cos x - e^x \sin x$$

at  $(0, 1)$

$$y' = e^0 \cos 0 - e^0 \sin 0$$
$$= (1) (1) - 0 = 1$$

∴ Equation of the tangent line :

$$y - y_0 = m (x - x_0)$$

$$y - 1 = 1 (x - 0)$$

∴ Equation of the normal line :

$$y - y_0 = m_n (x - x_0)$$

$$y - 1 = \frac{1}{-1} (x - 0)$$



## Text book question



For what values of  $x$  does the graph of  $f$  have a horizontal tangent?

$$y = e^x \cos x$$

استقلاً بستا + ثبت استقلاً

$$\begin{aligned} y' &= e^x (1) \cos x + e^x (-\sin x) \\ &= e^x \cos x - e^x \sin x \end{aligned}$$

∴ The tangent line is horizontal

$$y' = 0$$

$$e^x \cos x - e^x \sin x = 0$$

$$e^x [\cos x - \sin x] = 0$$

$$e^x = 0$$

None

$$\cos x - \sin x = 0$$

$$\cos x = \sin x$$

$$\div \cos x$$

$$1 = \frac{\sin x}{\cos x}$$

$$\tan x = 1$$

$$\Rightarrow \therefore x = \frac{\pi}{4} + n\pi$$

where  $n$  is integer



|     |                |
|-----|----------------|
| S   | A <sup>#</sup> |
| # E | C              |



### Text book questions



Prove that  $\frac{d}{dx}(\csc x) = -\csc x \cot x$

$$\text{L.H.S} = \frac{d}{dx}(\csc x)$$

$$= \frac{d}{dx} \frac{1}{\sin x}$$

$$= \frac{\cancel{\sin x}^0 \cdot 0 - (1) \cos x}{(\sin x)^2}$$

$$= \frac{-\cos x}{\sin x \sin x} = -\csc x \cot x = \text{R.H.S}$$

Prove that  $\frac{d}{dx}(\cot x) = -\csc^2 x$

$$\text{L.H.S} = \frac{d}{dx}(\cot x)$$

$$= \frac{d}{dx} \frac{1}{\tan x}$$

$$= \frac{\cancel{\tan x}^0 \cdot 0 - (1) \sec^2 x}{(\tan x)^2}$$

$$= \frac{-\frac{1}{\cancel{\sin^2 x}^{\cancel{\cos^2 x}}}}{\frac{\sin^2 x}{\cancel{\cos^2 x}}} = \frac{-1}{\sin^2 x} = -\csc^2 x = \text{R.H.S}$$



**For A Student !!**



**Text book question**



**Find the given derivative**

$$y = \sin x$$

$$\frac{d^{99}}{dx^{99}}(\sin x)$$

$$1) y' = \cos x$$

$$2) y'' = -\sin x$$

$$3) y''' = -\cos x$$

$$4) y^{(4)} = \sin x$$

$$\therefore \frac{d^{99}}{dx^{99}}(\sin x) = -\cos x$$

**The 27<sup>th</sup> derivative of the function  $f(x) = \cos x - \sin x$  is :**

- a)  $-\cos x + \sin x$       b)  $\cos x + \sin x$       c)  $-\cos x - \sin x$   
d)  $\cos x - \sin x$       e) None of the above

**Apr 2019**

$$y = \cos x - \sin x$$

$$1) y' = -\sin x - \cos x$$

$$2) y'' = -\cos x + \sin x$$

$$3) y''' = \sin x + \cos x$$

$$4) y^{(4)} = \cos x - \sin x$$

$$\therefore f^{(27)}(x) = \sin x + \cos x$$

(b)



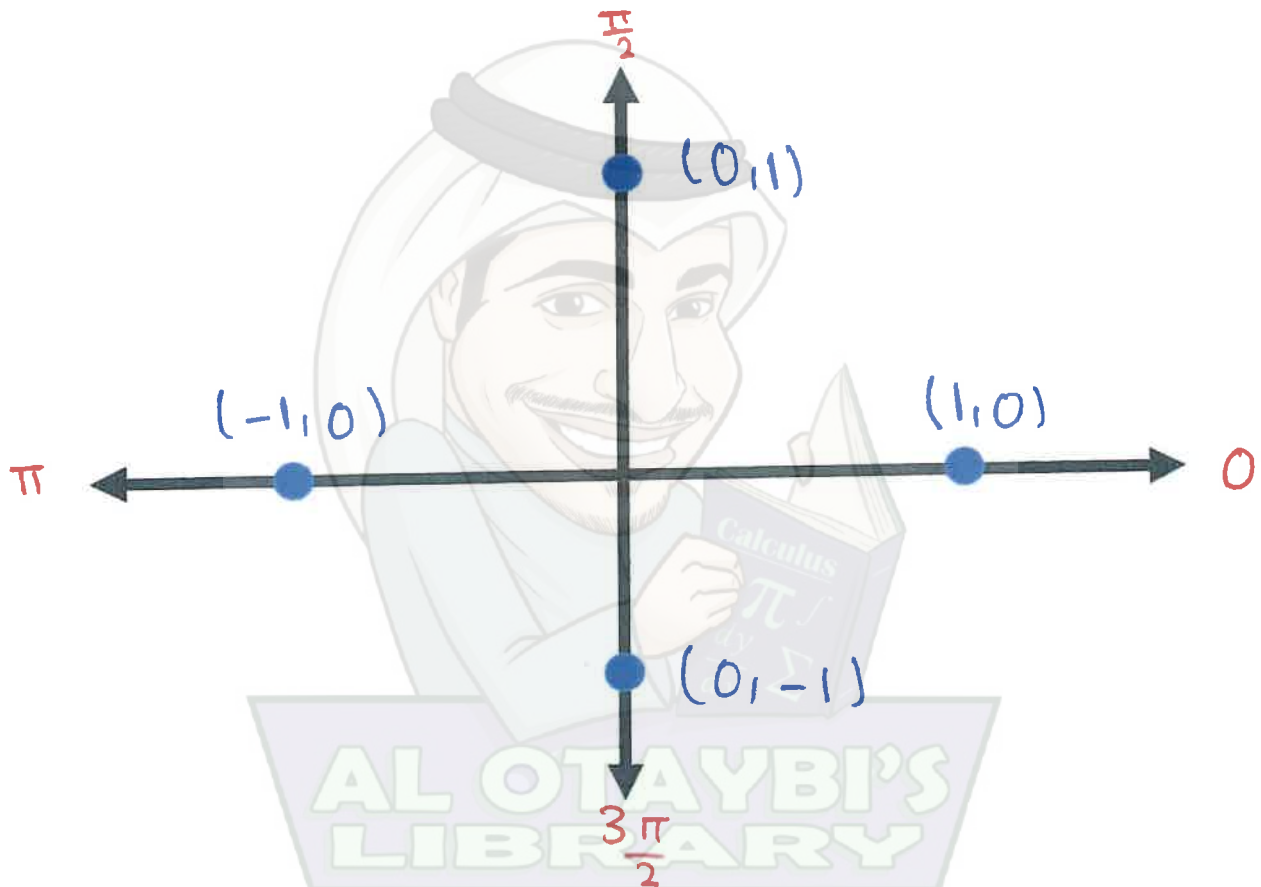
Review for angles  
مراجعة على الزوايا



الزوايا القائمة



$$P(x, y) = (\cos\theta, \sin\theta)$$



Find the value of

$$\sin(0) = 0$$

$$\tan(0) = \frac{\sin 0}{\cos 0} = \frac{0}{1} = 0$$

$$\cos(0) = 1$$

$$\sec(\pi) = \frac{1}{\cos \pi} = \frac{1}{-1} = -1$$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

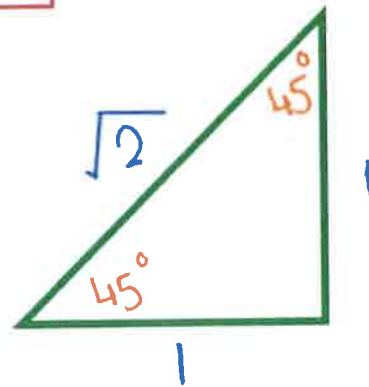
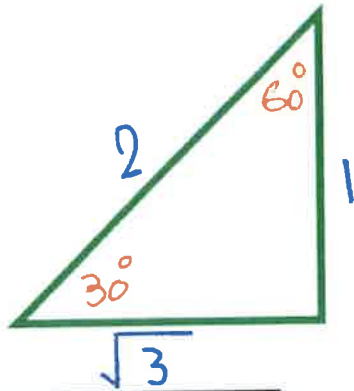
$$\csc\left(\frac{3\pi}{2}\right) = \frac{1}{\sin \frac{3\pi}{2}} = \frac{1}{-1} = -1$$



الزوايا  $45^\circ - 60^\circ - 30^\circ$



مثلثات



Find the value of

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

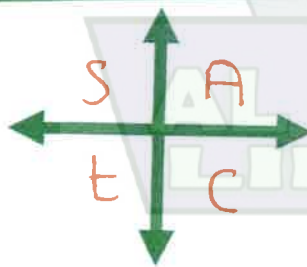
$$\tan\left(\frac{\pi}{4}\right) = \frac{1}{1} = 1$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\sec\left(\frac{\pi}{4}\right) = \frac{1}{\cos \frac{\pi}{4}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$



الزوايا الشاذة



الإشارة

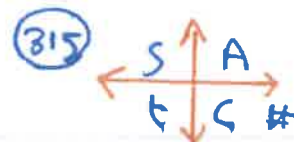
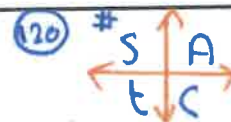
(1) أشوف الربع

(2) الزاوية =  $\frac{\pi}{\text{المقام}}$

Find the value of

$$\sin\left(\frac{2\pi}{3}\right) = + \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\sin\left(\frac{7\pi}{4}\right) = - \sin \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$



بعض قوانين الدوال المثلثية



$$\cos^2 x + \sin^2 x = 1$$

$$\div \cos^2 x$$

$$1 + \tan^2 x = \sec^2 x$$

$$\div \sin^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \begin{cases} \cos^2 x - \sin^2 x \\ 2\cos^2 x - 1 \\ 1 - 2\sin^2 x \end{cases}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

$$\sin(-x) = -\sin(x) \quad , \quad \tan(-x) = -\tan(x)$$
$$\cos(-x) = \cos(x)$$

cos

تبلغ السالب  
بزاويتها





## Techniques for finding limits of trigonometric function



$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$



ملاحظة حلوة



يمكن التبديل بين البسط والمقام

أول قانونين

يمكن التبديل بالترتيب في البسط

القانون الثالث

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$



## شروط تطبيق القوانين



- (1) الزاوية نفس المقام
- (2) اللمت تسعى لصفر الزاوية



True or false ( if false , find the correct answer )



$$\lim_{x \rightarrow 5} \frac{\sin(x-5)}{x-5} = 1$$



True

$$\lim_{x \rightarrow 3} \frac{3-x}{\tan(3-x)} = 1$$



True

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 1$$



False

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x} = 1$$



False

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x} = \frac{\sin \pi}{\pi} = \frac{0}{\pi} = 0$$

$$\lim_{x \rightarrow -1} \frac{\sin(x+1)}{x+1} = 1$$



True

$$\lim_{x \rightarrow 0} \frac{\sin(x+1)}{x+1} = 1$$



False

$$\lim_{x \rightarrow 0} \frac{\sin(x+1)}{x+1} = \frac{\sin 1}{1} = \sin 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{3 + \sin x}{2 + x}$$

$$= \frac{3 + \sin 0}{2 + 0} = \frac{3 + 0}{2} = \frac{3}{2}$$



Text book question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$$

$$\frac{0}{0}$$

Solution # 1

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\cancel{5x} \frac{\sin 5x}{\cancel{5x}}}{\cancel{3x}} \\ &= \lim_{x \rightarrow 0} \frac{5 \left( \frac{\sin 5x}{5x} \right)}{3} \\ &= \frac{5 (1)}{3} = \frac{5}{3} \end{aligned}$$

Solution # 2

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\sin 5x}{3x} \times \frac{5}{5} \\ &= \lim_{x \rightarrow 0} \frac{5 \sin 5x}{3 \cdot 5x} \\ &= \frac{5}{3} (1) = \frac{5}{3} \end{aligned}$$



## Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{5x^3 - 4x}$$

$$\frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{x(5x^2 - 4)}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{3x} \frac{\sin 3x}{3x}}{\cancel{x}(5x^2 - 4)} = \frac{3(1)}{0 - 4} = \frac{-3}{4}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{x^2}$$

$$\frac{0}{0}$$

**Solution # 1**

**Solution # 2**

$$= \lim_{x \rightarrow 0} \frac{\cancel{3x} \frac{\sin 3x}{3x} \frac{\sin 5x}{5x} \cancel{5x}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{x \cdot x} \frac{3(5)}{3(5)}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{15x^2} \left( \frac{\sin 3x}{3x} \right) \left( \frac{\sin 5x}{5x} \right)}{\cancel{x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{15 \left( \frac{\sin 3x}{(3)} \right) \left( \frac{\sin 5x}{(5)} \right)}{1}$$

$$= \frac{15(1)(1)}{1} = 15$$

$$= 15(1)(1) = 15$$

Find the limit if it exist

$$\lim_{x \rightarrow 0} \frac{3x + \tan 2x}{2x + 3\sin x}$$

$$\lim_{x \rightarrow 0} \frac{3x + \tan 2x}{2x + 3\sin x} \quad \frac{0 + \tan 0}{0 + 3\sin 0} \quad \frac{0+0}{0+0} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{3x + \cancel{2x} \frac{\tan 2x}{\cancel{2x}}}{2x + 3 \cancel{x} \frac{\sin x}{\cancel{x}}}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x} [3 + 2 \frac{\tan 2x}{\cancel{2x}}]}{\cancel{x} [2 + 3 \frac{\sin x}{\cancel{x}}]}$$

$$= \frac{3 + 2(1)}{2 + 3(1)} = \frac{5}{5} = 1$$

Big Mistake

$$\lim_{x \rightarrow 0} \frac{3x + \cancel{2x} \frac{\tan 2x}{\cancel{2x}}}{2x + 3 \cancel{x} \frac{\sin x}{\cancel{x}}}$$

$$\lim_{x \rightarrow 0} \frac{3x + 2x(1)}{2x + 3x(1)}$$

\* انتبه \*



## Text book questions



Evaluate the limit if it exist

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\sin \theta}$$

$$\frac{1-1}{0} = \frac{0}{0}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cancel{\theta} \frac{\cos \theta - 1}{\theta}}{\cancel{\theta} \frac{\sin \theta}{\theta}}$$

$$= \frac{0}{1} = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin x}{\sin \pi x}$$

$$\frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x} \frac{\sin x}{x}}{\cancel{\pi x} \frac{\sin \pi x}{\pi x}}$$

$$= \frac{1}{\pi (1)} = \frac{1}{\pi}$$

Find the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin(3x^2)}{7x^3 - 3x^2}$$

$$\frac{\sin 0}{0-0} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\sin(3x^2)}{x^2(7x-3)}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{3x^2} \frac{\sin(3x^2)}{3x^2}}{\cancel{x^2}(7x-3)} = \frac{3(1)}{0-3} = \frac{3}{-3} = -1$$

Find the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\tan^3 x}{x^4 + 5x^3}$$

$$\frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\tan^3 x}{x^3(x+5)}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x^3} \frac{\tan^3 x}{\cancel{x^3}}}{\cancel{x^3}(x+5)}$$

$$= \frac{1}{0+5} = \frac{1}{5}$$

S.O.L #2  
محلولة  
قانون  
= (1)<sup>3</sup>  $\frac{1}{0+5}$   
( $\frac{1}{5}$ )



## Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2 + x - 2}$$

$$\frac{\sin 0}{1+1-2} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x+2)}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)} \frac{\sin(x-1)}{x-1}}{(\cancel{x-1})(x+2)} = \frac{1}{3}$$

Evaluate the limit if it exist

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta + \tan \theta}$$

$$\frac{0}{0}$$

$$= \lim_{\theta \rightarrow 0} \frac{\theta \frac{\sin \theta}{\theta}}{\theta + \theta \frac{\tan \theta}{\theta}}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cancel{\theta} \frac{\sin \theta}{\theta}}{\cancel{\theta} \left[ 1 + \frac{\tan \theta}{\theta} \right]}$$

$$= \frac{1}{1+1} = \frac{1}{2}$$



## Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x}$$

$\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{x^2 \frac{\sin(x^2)}{x^2}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{x \left( \frac{\sin(x^2)}{x^2} \right)}{1}$$

$$= 0 (1) = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \csc x \sin(\sin x)$$

**Solution # 1**

$$= \lim_{x \rightarrow 0} \frac{1}{\sin x} \sin(\sin x)$$

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{\sin x} \left( \frac{\sin(\sin x)}{\cancel{\sin x}} \right)}{\cancel{\sin x}}$$

$$= 1$$

**Solution # 2**

$$= \lim_{x \rightarrow 0} \frac{1}{\sin x} \sin(\sin x)$$

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x}$$

$$= 1$$

Find the limit if it exist

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)\tan(x-1)}{x-1}$$

0/0

$$\lim_{x \rightarrow 1} \frac{\cancel{(x-1)} \frac{\sin(\cancel{x-1})}{\cancel{x-1}} \frac{\tan(\cancel{x-1})}{\cancel{x-1}}}{\cancel{x-1}}$$

$$= (1)(1)(1-1) = 0$$

Sol #2 بصرية

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)\tan(x-1)}{x-1}$$

لقد استخدمنا قانون

$$= (1)\tan 0 = 0$$

Find the limit if it exist

$$\lim_{x \rightarrow 0} \frac{x^2 + \sin^2(3x)}{5x^2}$$

0+0/0

$$\lim_{x \rightarrow 0} \frac{x^2 + \sin(3x)\sin(3x)}{5x^2}$$

$$\lim_{x \rightarrow 0} \frac{x^2 + 3x \frac{\sin 3x}{3x} \frac{\sin 3x}{3x} 3x}{5x^2}$$

$$\lim_{x \rightarrow 0} \frac{x^2 + 9x^2 \frac{\sin 3x}{3x} \frac{\sin 3x}{3x}}{5x^2}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x^2} [1 + 9 \frac{\sin 3x}{3x} \frac{\sin 3x}{3x}]}{5 \cancel{x^2}}$$

$$= \frac{1 + 9(1)(1)}{5} = \frac{10}{5} = 2$$

Sol #2

$$\lim_{x \rightarrow 0} \left( \frac{x^2}{5x^2} + \frac{\sin^2(3x)}{5x^2} \right)$$

$$\frac{1}{5} + \lim_{x \rightarrow 0} \frac{3^2 \sin^2(3x)}{3^2 5x^2}$$

$$\lim_{x \rightarrow 0} \frac{9}{5} \left( \frac{\sin 3x}{3x} \right)^2$$

$$\frac{1}{5} + \frac{9}{5} (1)^2 = \frac{10}{5} = 2$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow -2} \frac{\sin(x+2)}{x^3 + 8}$$

$$\frac{\sin 0}{0} \quad \frac{0}{0}$$

Sol #1

$$\lim_{x \rightarrow -2} \frac{\sin(x+2)}{(x+2)(x^2-2x+4)}$$

$$\lim_{x \rightarrow -2} \frac{\cancel{(x+2)} \frac{\sin(x+2)}{x+2}}{(x+2)(x^2-2x+4)}$$

$$= \frac{1}{4+4+4} = \frac{1}{12}$$

Sol #2

بسرعة

$$\lim_{x \rightarrow -2} \frac{\sin(x+2)}{(x+2)(x^2-2x+4)}$$

$$(1) \frac{1}{4+4+4} = \frac{1}{12}$$

Find the limit if it exist

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 - 3x + 2}$$

$$\frac{\sin 0}{0} \quad \frac{0}{0}$$

Sol #1

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-1)(x-2)}$$

$$\lim_{x \rightarrow 2} \frac{\cancel{(x-2)} \frac{\sin(x-2)}{x-2}}{(x-1)\cancel{(x-2)}}$$

$$= \frac{1}{2-1} = \frac{1}{1} = 1$$

Sol #2

بسرعة

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-1)(x-2)}$$

$$= \frac{1}{2-1} (1)$$

$$= (1)(1) = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{x - 2}$$

$$\frac{\sin 0}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - 4) \frac{\sin(x^2 - 4)}{x^2 - 4}}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x+2) \frac{\sin(x^2 - 4)}{x^2 - 4}}{\cancel{x-2}}$$

$$= \frac{(4)(1)}{1} = 4$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 2} (x - 2)^2 \cot^2(x - 2)$$

$$= \lim_{x \rightarrow 2} (x - 2)^2 \frac{1}{\tan^2(x - 2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x - 2)^2}{\tan^2(x - 2)} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}^2}{\cancel{(x-2)}^2 \tan^2(x-2)} = \frac{1}{1} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sin x} - 1}{\tan x}$$

$$\frac{\sqrt{1+0} - 1}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sin x} - 1}{\tan x} * \frac{\sqrt{1 + \sin x} + 1}{\sqrt{1 + \sin x} + 1}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x} + \sin x - \cancel{x}}{\tan x [\sqrt{1 + \sin x} + 1]}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x} \frac{\sin x}{\cancel{x}}}{\cancel{x} \frac{\tan x}{\cancel{x}} [\sqrt{1 + \sin x} + 1]}$$

$$= \frac{1}{1 [\sqrt{1+0} + 1]}$$

$$= \frac{1}{1+1} = \frac{1}{2}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{|x|}{\sin 2x}$$

$$\lim_{x \rightarrow 0^+} \frac{x}{\sin 2x} \quad 0/0$$

$$\lim_{x \rightarrow 0^+} \frac{x}{2x \cdot \frac{\sin 2x}{2x}}$$

$$= \frac{1}{2(1)} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0^-} \frac{-x}{\sin 2x}$$

$$\lim_{x \rightarrow 0^-} \frac{-x}{2x \cdot \frac{\sin 2x}{2x}}$$

$$= \frac{-1}{2(1)} = -\frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

أفكار  
لبسرة

$$\lim_{x \rightarrow 0^+} \frac{(2x)}{2(\sin 2x)} = \frac{1}{2} (1) = \frac{1}{2}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x^3 + x^2}}$$

0/0

$$\lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x^2(x+1)}}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{|x| \sqrt{x+1}}$$

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x \sqrt{x+1}}$$

$$\lim_{x \rightarrow 0^-} \frac{\sin x}{-x \sqrt{x+1}}$$

$$\lim_{x \rightarrow 0^+} \frac{\cancel{x} \frac{\sin x}{x}}{\cancel{x} \sqrt{x+1}}$$

$$\lim_{x \rightarrow 0^-} \frac{\cancel{-x} \frac{\sin x}{x}}{\cancel{-x} \sqrt{x+1}}$$

أقبل  
على  
أعلى  
الفتحة

$$= \frac{1}{\sqrt{0+1}}$$

$$= \frac{1}{1} = 1$$

$$= \frac{1}{-\sqrt{0+1}}$$

$$= \frac{1}{-1} = -1$$

$$\therefore \lim_{x \rightarrow 0^+} \frac{\sin x}{\sqrt{x^3 + x^2}} \neq \lim_{x \rightarrow 0^-} \frac{\sin x}{\sqrt{x^3 + x^2}}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x^3 + x^2}} \text{ DNE}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \sin x \cdot \sqrt{1 + \frac{1}{x^2}}$$

$$\begin{array}{l} 0 \sqrt{1 + \frac{1}{0}} \\ 0 \sqrt{1 + \infty} \\ 0 \sqrt{\infty} \\ 0 \infty \end{array}$$

$$\lim_{x \rightarrow 0} \sin x \sqrt{\frac{x^2}{x^2} + \frac{1}{x^2}}$$

$$\lim_{x \rightarrow 0} \sin x \sqrt{\frac{x^2 + 1}{x^2}}$$

$$\lim_{x \rightarrow 0} \sin x \frac{\sqrt{x^2 + 1}}{\sqrt{x^2}}$$

$$\lim_{x \rightarrow 0} \sin x \frac{\sqrt{x^2 + 1}}{|x|}$$

$$\lim_{x \rightarrow 0^+} \sin x \frac{\sqrt{x^2 + 1}}{x}$$

$$\lim_{x \rightarrow 0^-} \sin x \frac{\sqrt{x^2 + 1}}{-x}$$

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x} \cdot \frac{\sqrt{x^2 + 1}}{1}$$

$$\lim_{x \rightarrow 0^-} \frac{\sin x}{-x} \cdot \frac{\sqrt{x^2 + 1}}{1}$$

$$= (1) \sqrt{0+1} = 1$$

$$= (1) \frac{\sqrt{0+1}}{-1} = -1$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{x^2}$$

0/0 0/0

$$\lim_{x \rightarrow 0} \frac{\sin x [1 - \cos x]}{x^2}$$

Sol #1

$$\lim_{x \rightarrow 0} \frac{x \frac{\sin x}{x} \frac{(1 - \cos x)}{x} x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{x^2 \frac{\sin x}{x} \frac{1 - \cos x}{x}}{x^2}$$

$$= (1)(0) = 0$$

Sol #2

للسرعة

$$\lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{x \cdot x}$$

$$= (1)(0) = 0$$

AL OTAYBI'S  
LIBRARY



## Tricky question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin 2x - 2 \sin x}{x^2}$$

$\frac{0-0}{0} \quad \frac{0}{0}$

$$\lim_{x \rightarrow 0} \frac{2 \sin x \cos x - 2 \sin x}{x^2}$$

**Remark**

$$\sin 2x = 2 \sin x \cdot \cos x$$



$$\lim_{x \rightarrow 0} \frac{2 \sin x [\cos x - 1]}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \times \frac{\sin x}{x} \times \frac{(\cos x - 1)}{x} \times x}{x^2}$$

SOL H2

بسهولة

$$\lim_{x \rightarrow 0} \frac{2 \cancel{x^2} \times \frac{\sin x}{x} \times \frac{\cos x - 1}{x}}{\cancel{x^2}}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x [\cos x - 1]}{x \cdot x}$$

$$= 2 (1) (0)$$

$$= 2 (1) (0) = 0$$

$$= 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$$

$\infty \sin \frac{1}{\infty}$   
 $\infty \sin 0$   
 $\infty \cdot 0$

$$\lim_{x \rightarrow \infty} x \times \frac{1}{x} \times \sin \frac{1}{x}$$

$$= 1$$



Nice question !



Text book questions



Evaluate the limit if it exist

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{2\theta^2}$$

$$\frac{1-1}{0} \quad \frac{0}{0}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{2\theta^2} * \frac{\cos \theta + 1}{\cos \theta + 1}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cos^2 \theta - 1}{2\theta^2 (\cos \theta + 1)} \rightarrow \begin{aligned} \cos^2 \theta + \sin^2 \theta &= 1 \\ \cos^2 \theta - 1 &= -\sin^2 \theta \end{aligned}$$

$$= \lim_{\theta \rightarrow 0} \frac{-\sin^2 \theta}{2\theta^2 (\cos \theta + 1)}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cancel{-\theta^2} \frac{\sin^2 \theta}{\theta^2}}{\cancel{2\theta^2} (\cos \theta + 1)}$$

$$= \frac{-1}{2(1+1)} = \frac{-1}{4}$$

Past exam  
Nov 2018

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{3\theta^2}$$



Nice question !



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{1 - \cos(x - 1)}{(x - 1)^2}$$

$$\frac{1-1}{0} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{1 - \cos(x-1)}{(x-1)^2} * \frac{1 + \cos(x-1)}{1 + \cos(x-1)}$$

$$= \lim_{x \rightarrow 1} \frac{1 - \cos^2(x-1)}{(x-1)^2 [1 + \cos(x-1)]} \rightarrow \begin{aligned} \cos^2(x-1) + \sin^2(x-1) &= 1 \\ \sin^2(x-1) &= 1 - \cos^2(x-1) \end{aligned}$$

$$= \lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{(x-1)^2 [1 + \cos(x-1)]}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)^2} \sin^2(x-1)}{\cancel{(x-1)^2} [1 + \cos(x-1)]}$$

$$= \frac{1}{1 + \cos 0} = \frac{1}{1 + 1} = \frac{1}{2}$$



Nice questions !



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x \sin x}$$

$$\frac{1-1}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{x \sin x} * \frac{\cos x + 1}{\cos x + 1}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x \sin x (\cos x + 1)} \xrightarrow{\cos^2 x + \sin^2 x = 1} \cos^2 x - 1 = -\sin^2 x$$

$$= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x \sin x (\cos x + 1)}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{x (\cos x + 1)}$$

$$= \lim_{x \rightarrow 0} \frac{-x \left( \frac{\sin x}{x} \right)}{x (\cos x + 1)}$$

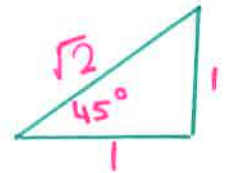
$$= \frac{-1}{\cos 0 + 1} = \frac{-1}{1 + 1} = \frac{-1}{2}$$



Tricky question !!



Text book question



Evaluate the limit if it exist

$$\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{\sin x - \cos x}$$

$$\frac{1 - 1}{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}} = \frac{0}{0}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow \pi/4} \frac{1 - \frac{\sin x}{\cos x}}{\sin x - \cos x} \\
 &= \lim_{x \rightarrow \pi/4} \frac{\frac{\cos x}{\cos x} - \frac{\sin x}{\cos x}}{\sin x - \cos x} \\
 &= \lim_{x \rightarrow \pi/4} \frac{\cos x - \sin x}{\cos x (\sin x - \cos x)} \\
 &= \lim_{x \rightarrow \pi/4} \frac{\cos x - \sin x}{\cos x (\sin x - \cos x)} \\
 &= \lim_{x \rightarrow \pi/4} \frac{- (\cancel{\sin x} - \cos x)}{\cos x (\cancel{\sin x} - \cos x)} \\
 &= \frac{1}{\frac{1}{\sqrt{2}}} = -\sqrt{2}
 \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\sin(x^2) + 2x}{x}$$

Apr 2019

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{x^2 \frac{\sin x^2}{x^2} + 2x}{x} \\ &= \lim_{x \rightarrow 0} \frac{\cancel{x} \left[ \cancel{x} \frac{\sin x^2}{x^2} + 2 \right]}{\cancel{x}} = 0(1) + 2 = 0 + 2 = 2 \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 + x - 6}$$

Dec 2016

$$\begin{aligned} &= \lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+3)} \\ &= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)} \frac{\sin(x-2)}{\cancel{x-2}}}{\cancel{(x-2)} (x+3)} = \frac{1}{5} \end{aligned}$$

$$\lim_{x \rightarrow 0} x \cot(\pi x) =$$

July 2017

a)  $\pi$    b)  $-\pi$    c)  $\frac{1}{\pi}$    d)  $-\frac{1}{\pi}$    e) None of the above

$$\begin{aligned} &= \lim_{x \rightarrow 0} x \frac{1}{\tan(\pi x)} = \lim_{x \rightarrow 0} \frac{x}{\tan(\pi x)} \\ &= \lim_{x \rightarrow 0} \frac{\cancel{x}}{\pi \cancel{x} \frac{\tan(\pi x)}{\pi x}} = \frac{1}{\pi(1)} = \frac{1}{\pi} \quad (c) \end{aligned}$$

### 3.4 The chain rule



#### Remark

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$$

$$(a^x)' = a^x \ln a$$

$$(a^{f(x)})' = a^{f(x)} f'(x) \ln a$$

Differentiate the function

$$f(x) = 5^x$$

$$f'(x) = 5^x (\ln 5)$$

$$f(x) = 7^{\frac{1}{x}}$$

$$f'(x) = 7^{\frac{1}{x}} \left(-\frac{1}{x^2}\right) \ln 7$$

$$f(x) = 10^{x^2}$$

$$f'(x) = 10^{x^2} \cdot 2x \ln 10$$



Text book questions



Find the derivative of the function

$$F(x) = (5x^6 + 2x^3)^4$$

$$F'(x) = 4(5x^6 + 2x^3)^3 (30x^5 + 6x^2)$$

$$F(x) = (1 + x + x^3)^{99}$$

$$F'(x) = 99(1 + x + x^3)^{98} (1 + 3x^2)$$



## Text book questions



Find the derivative of the function

$$f(t) = t \sin \pi t$$

اشتقاق بشتا + بشتا اشتقاق

$$f'(t) = (1) \sin \pi t + t \cos \pi t \cdot \pi$$

$$F(t) = e^{at} \sin bt$$

اشتقاق بشتا + بشتا اشتقاق

$$F'(t) = e^{at} \cdot a \sin bt + e^{at} \cos bt \cdot b$$

$$F(t) = \tan(e^t) + e^{\tan t}$$

$$F'(t) = \sec^2(e^t) e^t (1) + e^{\tan t} \cdot \sec^2 t (1)$$

$$F(x) = e^{\sin 2x} + \sin(e^x)$$

$$F'(x) = e^{\sin 2x} \cdot \cos 2x \cdot 2 + \cos(e^x) \cdot e^x (1)$$

$$y = \sec^2(m\theta) , \text{ where } m \text{ is a constant}$$

$$y' = 2 \sec(m\theta) \sec(m\theta) \cdot \tan(m\theta) \cdot m$$

Find  $f'(x)$  where

$$f(x) = \sin^2(1 + \sqrt{x})$$

انتبه !!  
أس !



$$f(x) = [\sin(1 + \sqrt{x})]^2$$
$$f'(x) = 2 [\sin(1 + \sqrt{x})] \cos(1 + \sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$

Find  $y'$  where

$$y = \left( \tan\left(\frac{x}{2}\right) + \cos 2x \right)^3$$

$$y' = 3 \left( \tan\left(\frac{x}{2}\right) + \cos 2x \right)^2 \left( \sec^2\left(\frac{x}{2}\right) \cdot \frac{1}{2} - \sin 2x \cdot 2 \right)$$

Find the derivative of the function

$$y = e^{-2t} \cos 4t$$

الشتق بـ  $t$  + بـ  $t$  الشقاق

انتبه !!  
ضرب !



$$y' = e^{-2t} (-2) \cos 4t + e^{-2t} (-\sin 4t \cdot 4)$$

Find the derivative of

$$f(t) = e^{t \sin 2t}$$

$$f'(t) = e^{t \sin 2t} [ (1) \sin 2t + t \cos 2t \cdot 2 ]$$

الشتق بـ  $t$  + بـ  $\sin 2t$

Find  $y'$  where

$$y = \sin(e^t) + e^{\sin t}$$

$$y' = \cos(e^t) e^t (1) + e^{\sin t} \cos t$$

Find  $f'(x)$  where

$$f(x) = \frac{\sec x}{x^3 + \cot x}$$

انتبه !!  
بسط ومقام

$$f'(x) = \frac{(x^3 + \cot x) \sec x \tan x - \sec x (3x^2 - \csc^2 x)}{(x^3 + \cot x)^2}$$



## Text book questions



Find the derivative of the function

$$y = \sqrt{\frac{x}{x+1}}$$

$$y' = \frac{1}{2 \sqrt{\frac{x}{x+1}}} \cdot \frac{(x+1)(1) - x(1)}{(x+1)^2}$$

$$y = \sqrt{\frac{1 + \sin t}{1 + \cos t}}$$

$$y' = \frac{1}{2 \sqrt{\frac{1 + \sin t}{1 + \cos t}}} \cdot \frac{(1 + \cos t) \cos t - (1 + \sin t)(-\sin t)}{(1 + \cos t)^2}$$

$$g(u) = \left( \frac{u^3 - 1}{u^3 + 1} \right)^8$$

$$g'(u) = 8 \left( \frac{u^3 - 1}{u^3 + 1} \right)^7 \cdot \frac{(u^3 + 1) 3u^2 - (u^3 - 1) 3u^2}{(u^3 + 1)^2}$$

$$g(u) = \frac{(r^2 - 1)^2}{(2r + 1)^5}$$

$$g'(u) = \frac{(2r+1)^5 \cdot 2(r^2-1) \cdot 2r - (r^2-1)^2 \cdot 5(2r+1)^4 \cdot 2}{[(2r+1)^5]^2}$$

Find the derivative of the function

$$y = (x^4 + 3x^2 - 5)^5 + \sqrt{1 - 2x}$$

$$y' = 5(x^4 + 3x^2 - 5)^4 (4x^3 + 6x) + \frac{(-2)}{2\sqrt{1-2x}}$$

$$f(\theta) = e^{\sec 3\theta} + 3^{\tan \theta}$$

$$f'(\theta) = e^{\sec 3\theta} \cdot \sec 3\theta \cdot \tan 3\theta (3) + 3^{\tan \theta} \cdot \sec^2 \theta \ln 3$$

$$f(x) = a^3 + \cos^3 x + xe^{-kx}$$

$$f'(x) = 0 + 3\cos^2 x (-\sin x) + (1)e^{-kx} + xe^{-kx} (-k)$$

استقنا بثت + بثت استقنا



Homework



$$y = (e^{4x} + 7^{2x})^3 + \sin(2^x + 1)$$

$$y' = 3(e^{4x} + 7^{2x})^2 (e^{4x} (4) + 7^{2x} (2) \ln 7 + \cos(2^x + 1) 2^x \ln 2)$$



## Text book questions



Find the derivative of the function

$$f(z) = e^{\frac{z}{z-1}}$$

$$f'(z) = e^{\frac{z}{z-1}} \cdot \frac{(z-1)(1) - z(1)}{(z-1)^2}$$

$$F(t) = e^{r \sin 2t}$$

$$F'(t) = e^{r \sin 2t} \left[ \overset{\text{تابع}}{r} \cos 2t \cdot (2) \right]$$

$$r(t) = 10^{2\sqrt{t}}$$

$$r'(t) = 10^{2\sqrt{t}} \cdot \frac{1}{2\sqrt{t}} \cdot \ln 10$$

$$r(t) = 2^{t^3}$$

$$r'(t) = 2^{t^3} \cdot 3t^2 \cdot \ln 2$$

$$r(t) = 4^{C/x} \text{ where } C \text{ is constant}$$

$$r'(t) = 4^{\frac{C}{x}} \cdot \frac{-C}{x^2} \cdot \ln 4$$



Tricky questions !!



Text book questions



Find the derivative of the function

$$f(x) = (2x - 4)^4 (x^2 + x + 1)^5$$

استقنا بـ ٤ + استقنا بـ ٥

$$f'(x) = 4(2x-4)^3 (2)(x^2+x+1)^5 + (2x-4)^4 \cdot 5(x^2+x+1)^4 (2x+1)$$

$$h(t) = (t + 1)^{3/2} (2t^2 - 1)^3$$

استقنا بـ ٣/٢ + استقنا بـ ٣

$$h'(t) = \frac{3}{2}(t+1)^{\frac{1}{2}} (1)(2t^2-1)^3 + (t+1)^{\frac{3}{2}} 3(2t^2-1)^2 4t$$

$$y = \cos \sqrt{\sin(\tan \pi x)}$$

$$y' = -\sin \sqrt{\sin(\tan \pi x)}$$

$$\frac{\cos(\tan \pi x) \sec^2 \pi x \cdot \pi}{2 \sqrt{\sin(\tan \pi x)}}$$



Tricky questions !!



Text book questions



Find the derivative of the function

$$y = [x + (x + \sin^2 x)^3]^5$$

$$y' = 5 [x + (x + \sin^2 x)^3]^4 [1 + 3(x + \sin^2 x)^2 [1 + 2 \sin x \cos x]]$$

$$f(x) = \left( \frac{e^x}{x+1} \right)^7 \left( \frac{x^2}{e^{2x}-1} \right)^9$$

استقلاً بـ + بـ استقلاً

$$f'(x) = 7 \left( \frac{e^x}{x+1} \right)^6 \frac{(x+1)e^x - e^x(1)}{(x+1)^2} \left( \frac{x^2}{e^{2x}-1} \right)^9$$

$$+ \left( \frac{e^x}{x+1} \right)^7 \cdot 9 \left( \frac{x^2}{e^{2x}-1} \right)^8 \frac{(e^{2x}-1)2x - x^2 e^{2x}(2)}{(e^{2x}-1)^2}$$

If  $F(x) = 3f(4f(x))$ , where  $f(0) = 0$  and  $f'(0) = 2$ ,  
then  $F'(0)$  is equal to

a)48   b)32   c)16   d)0   e)None of the above

Apr 2019

$$F(x) = 3 f(4f(x))$$

$$F'(x) = 3 f'(4f(x)) \cdot 4 f'(x)$$

at  $x=$

$$F'(0) = 3 f'(4f(0)) \cdot 4 f'(0)$$

$$= 3 f'(4(0)) \cdot 4 (2)$$

$$= 3 f'(0) \cdot 8$$

$$= 3 (2) \cdot 8$$

$$= 6 \cdot (8)$$

$$= 48 \quad \text{(a)}$$



### Text book question



Find all points on the graph of the function

$f(x) = 2\sin x + \sin^2 x$  at which the tangent line is horizontal.

$$f'(x) = 2\cos x + 2\sin x \cos x$$

$\therefore$  The tangent line is horizontal

$$\therefore f'(x) = 0$$

$$2\cos x + 2\sin x \cos x = 0$$

$$2\cos x [1 + \sin x] = 0$$

$$2\cos x = 0$$

$\div 2$

$$\cos x = 0$$

$$1 + \sin x = 0$$

$$\sin x = -1$$

$$x = \frac{\pi}{2} + 2n\pi$$

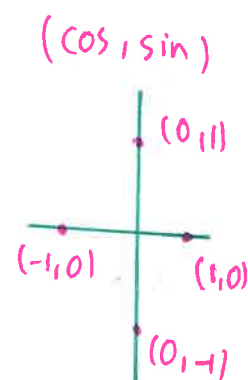
where  $n$  is  $\pm$  integer

$$\text{at } x = \frac{\pi}{2} + 2n\pi$$

$$y = 2(1) + (1)^2 = 3 \implies P\left(\frac{\pi}{2} + 2n\pi, 3\right)$$

$$\text{at } x = \frac{3\pi}{2} + 2n\pi$$

$$y = 2(-1) + (-1)^2 = -2 + 1 = -1 \implies P\left(\frac{3\pi}{2} + 2n\pi, -1\right)$$





### Text book question



At what point on the curve  $y = \sqrt{1+2x}$  is the tangent line perpendicular to the line  $6x + 2y = 1$ ?

$$y = \sqrt{1+2x}$$

$$y' = \frac{2}{2\sqrt{1+2x}}$$

$$6x + 2y = 1$$

$$2y = 1 - 6x$$

$$\div 2$$

$$y = \frac{1}{2}(1 - 6x)$$

$$y = \frac{1}{2} - 3x$$

$$y'_2 = -3$$

$\therefore$  The tangent line is perpendicular to the line.

$$\therefore y'_1 = \frac{-1}{y'_2}$$

$$\frac{1}{\sqrt{1+2x}} = \frac{1}{3}$$

$$3 = \sqrt{1+2x}$$

$$9 = 1 + 2x$$

$$\longrightarrow 8 = 2x \longrightarrow x = 4$$

$$\text{at } x = 4 \longrightarrow y = \sqrt{1+8} = \sqrt{9} = 3$$

$$\Rightarrow P(4, 3)$$



## Text book questions



Identify the inner function  $u = g(x)$  and the outer function  $y = f(u)$ . Then find  $\frac{dy}{dx}$

$$y = \tan(\sin x)$$

inner function  $\rightarrow u = \sin x$

outer function  $\rightarrow y = \tan u$

$$y' = \sec^2(\sin x) \cos x$$

$$y = e^{\sqrt{x}}$$

inner function  $\rightarrow u = \sqrt{x}$

outer function  $\rightarrow y = e^u$

$$y' = e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$$

$$y = (1 - x^2)^{10}$$

inner function  $\rightarrow u = 1 - x^2$

outer function  $\rightarrow y = u^{10}$

$$y' = 10(1 - x^2)^9 (-2x)$$

Let  $y = u^2 - 5u + 1$ ,  $u = \frac{x-1}{x+1}$

Find  $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= (2u - 5) \cdot \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2}$$

$$= \left( 2 \frac{x-1}{x+1} - 5 \right) \frac{x+1 - (x-1)}{(x+1)^2}$$

Find  $\frac{dy}{dx}$

$y = (u^2 + u + 1)^2$ ,  $u = \sqrt[3]{x \sec x + 1}$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= 2(u^2 + u + 1)(2u + 1) \cdot \frac{1}{3} (x \sec x + 1)^{-\frac{2}{3}} [ (1) \sec x + x \sec x \tan x ]$$

$$= 2 \left( (\sqrt[3]{x \sec x + 1})^2 + \sqrt[3]{x \sec x + 1} + 1 \right) (2\sqrt[3]{x \sec x + 1} + 1) \frac{1}{3} (x \sec x + 1)^{-\frac{2}{3}} [ \sec x + x \sec x \tan x ]$$

$$\text{Let } y = \left( \frac{u-1}{u+1} \right)^3 \text{ and } u = \sqrt[3]{2-3t-t^2}$$

$$= (2-3t-t^2)^{\frac{1}{3}}$$

Find  $\frac{dy}{dt}$  at  $t=2$

$$\text{at } t=2 \longrightarrow u = \sqrt[3]{2-6-4} = \sqrt[3]{-8} = -2$$

$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dt}$$

$$= 3 \left( \frac{u-1}{u+1} \right)^2 \frac{(u+1)(1) - (u-1)(1)}{(u+1)^2}$$

$$\cdot \frac{1}{3} (2-3t-t^2)^{-\frac{2}{3}} (-3-2t)$$

$$\text{at } t=2, u=-2$$

$$\frac{dy}{dt} = 3 \left( \frac{-3}{-1} \right)^2 \frac{-1 - (-3)}{(-1)^2} \cdot \frac{1}{3} (2-6-4)^{-\frac{2}{3}} (-3-4)$$

$$= \cancel{3} (9) \frac{-2}{1} \cancel{\frac{1}{3}} \frac{1}{(-8)^{\frac{2}{3}}} (-7)$$

$$\rightarrow (-2)^{\frac{2}{3}} = (-2)^2 = 4$$

$$= \cancel{9} \frac{1}{4} (-7)$$

$$= \frac{-63}{2}$$